



Society of Petroleum Engineers

SPE-229196-MS

Deep Learning Based Inpainting and Fault Segmentation of Fmi Logs Using Synthetic Deformation Images

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229196-MS](https://doi.org/10.2118/229196-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Formation micro-imagers record high-density resistivity along the borehole wall and unwrap it into a panorama that reveals bedding, fractures, vugs and other structural features at centimeter scale. Because pad electrodes cannot cover the entire circumference, final images invariably contain blank stripes that break lateral continuity and hinder both manual interpretation and automated analysis. This paper presents a comprehensive workflow for reconstructing FMI logs with realistic acquisition gaps and simultaneously identifying fault traces by combining synthetic data generation, acquisition mask transplantation and a context-aware convolutional network. We build FMI-like scenes from a simple stratigraphic template and subject them to smooth deformation, planar tilt, discrete faulting and unconformity truncation to produce diverse yet analytically labelled images. These two-dimensional realisations are extruded to three dimensions, unwrapped into a 360° borehole frame and wrapped back to a cylinder for quality control. To emulate field conditions, idealised missing-data masks are replaced with masks segmented from real FMI logs and randomly applied to the synthetic unwrapped images. The resulting inputs exhibit gap widths, azimuthal persistence and spacing consistent with the tool coverage in practice. A convolutional encoder–decoder ingests the masked image, a binary hole indicator and a fixed edge-magnitude channel. It is trained with a pixelwise reconstruction loss augmented by a lightly weighted adversarial term and a structural similarity term, thereby balancing structural fidelity with textural realism. Qualitatively the network restores sinusoidal bedding across occlusions, reconnects fault traces and maintains truncation geometry; quantitatively it converges stably despite non–power-of-two image sizes and improves the structural similarity index by six points over partial-convolution baselines. Realistic masks reduce seam artefacts and prevent overfitting to synthetic occlusion patterns, while explicit edge cues and hole awareness act synergistically to sharpen contacts and stabilise reconstruction. Finally, by sharing the encoder between inpainting and segmentation heads, we train a second branch to detect faults and achieve a mean intersection-over-union of 0.73 on unseen FMI slices. The results suggest that pairing controlled, analytically labelled deformation images with authentic acquisition gaps and context-aware neural architectures is an effective recipe for FMI inpainting and fault-aware interpretation.

Introduction

Fullbore Formation MicroImager (FMI) tools are pad-type micro-resistivity imagers that convert hundreds or thousands of voltage measurements per metre into an unwrapped two-dimensional panorama of the borehole. The resulting micro-resistivity image preserves centimetre-scale variation tied to lithology, bedding, fractures, vugs, sedimentary structures and drilling-induced artefacts. These images, which are provided in both dynamic and static colour scales, are among the richest wellbore datasets for structural geology and reservoir characterisation. The dynamic scale enhances local contrast to emphasise fine textures, while the static scale allows direct comparison of resistivity magnitudes across depth. Analysts routinely examine both modes in tandem to interpret bedding dips, identify fractures and faults, delineate thin beds, infer sedimentary facies and map borehole breakouts or drilling-induced fractures. FMI interpretation therefore contributes to exploration by constraining depositional environments and structural frameworks; to development planning by guiding well trajectories and completion design; and to operations by informing cement and packer placement and diagnosing production anomalies.

Despite their value, FMI logs are incomplete by design. Imaging pads and button electrodes cannot contact the entire borehole wall at all times because of tool geometry, eccentricity, caliper variations, washouts and standoff effects. The unmeasured sectors are rendered as blank stripes—often white bands—in the final mosaic. These blanks disrupt the continuity of bedding and structural features, complicating both manual interpretation and algorithmic analyses such as dip picking, fracture network characterisation and facies mapping. Overcoming this limitation has been a longstanding goal in electrical imaging, motivating a range of gap-filling or inpainting techniques. Beyond coverage, real FMI images are delivered in two modalities, dynamic and static, whose combined use aids quality control, structural interpretation and reservoir studies. The dynamic image highlights subtle textural variations, while the static image preserves absolute resistivity values; therefore any restoration method must respect both local texture and global intensity scaling.

Early gap-filling methods transplanted ideas from classical interpolation and geostatistics. Structure-based schemes, including inverse-distance and Lagrange interpolation, propagate values from neighbouring electrodes into blanks. They are simple and computationally efficient but have limited structural fidelity: seams are visible at strip edges and texture continuity suffers in complex geology. Multipoint geostatistics improves realism by copying texture patches from the unmasked regions into the target blanks. Filtersim and related template-matching algorithms paste the most similar patterns into the missing sectors, achieving better texture continuity and computational efficiency than naïve interpolation. Nevertheless, performance deteriorates in multilayered or strongly structured settings because sequential simulation and random choices can disrupt bedding continuity and bias the reconstructed textures. These limitations, especially in carbonates and conglomerates where textures are intricate, pushed the community towards learning-based approaches that can better capture multi-scale structure and texture.

Convolutional neural networks (CNNs) soon became the backbone of FMI inpainting. Plain encoder-decoder architectures reconstruct missing bands but leave obvious seams and lose fine texture. The U-Net architecture improves continuity by adding skip connections from encoder to decoder, preserving spatial detail and long-range context simultaneously. Attention-augmented U-Nets further constrain feature extraction to geologically salient regions and yield additional gains by focusing the model's capacity on relevant structures. Such FMI-specific baselines established that general-purpose inpainting architectures can restore missing strips to a degree, yet they still underperform in strongly textured intervals and may leave edge artefacts at strip boundaries.

A more powerful class of methods uses generative adversarial networks (GANs) to synthesise geologically plausible textures while maintaining global coherence. [Sun et al. \(2023\)](#) propose a ResGAN-U-Net with a multiscale generator and dual discriminators—a global discriminator enforcing whole-image consistency and a local discriminator focusing on patch-level details near strip edges. In their

implementation the encoder–decoder replaces max-pool down-sampling with strided convolutions, inserts residual blocks to ease optimisation and uses dilated convolutions to enlarge receptive fields without sacrificing resolution. Training mixes pixelwise mean-squared error with a modest adversarial term, yielding improvements in structural similarity index (SSIM), peak signal-to-noise ratio (PSNR) and lower Fréchet Inception Distance (FID) relative to encoder–decoder and U-Net baselines. Although the method produces sharper textures and fewer seam artefacts, its generator is heavy and the model can hallucinate textures not present in the training set if the adversarial loss dominates.

Self-supervised restoration techniques address the scarcity of curated training data. Deep Image Prior, introduced by Ulyanov et al. (2018), shows that the structure of an untrained CNN itself encodes strong natural-image statistics; optimising a randomly initialised network to match the known pixels of a single corrupted image can reconstruct the missing regions without any external training data. Regularisation by Denoising (RED) incorporates a denoiser prior into the optimisation as an explicit regulariser, making deblurring feasible when paired datasets are unavailable (Romano et al., 2017). Samarkin et al. (2023) combine Deep Image Prior with RED to inpaint and deblur FMI logs and outcrop analogues. Their approach requires no training corpus and works reasonably well on synthetic examples, but it is computationally expensive and tends to produce smooth textures compared with learning-based models. Meanwhile, Hou et al. (2023) propose a dual-modal transformer that operates directly on the dynamic and static FMI images and includes a "less-information dropping" module that removes blank bands before lithology classification. Their model attains approximately 90 percent accuracy on a field case in the Western Qaidam Basin. This work illustrates a complementary design choice: rather than filling the gaps for general interpretability, one can design task-specific networks that ignore the blanks. Nevertheless, when the goal is structural interpretation, gap filling remains essential.

More recently, free-form inpainting methods from computer vision have been adapted to geophysical images. Partial convolutional U-Nets restrict each convolution to the valid pixels within its receptive field and update the mask at every layer to mark locations as filled. Liu et al. (2018) demonstrate that this rule-based approach reduces artefacts and improves results on irregular masks compared with standard encoder–decoders. The U-Net topology preserves local structure while allowing deep layers to accumulate long-range information. The main limitation is that the mask update is heuristic: once a location is declared valid the network can no longer distinguish between observed and synthesised values, which may lead to oversmoothing near the boundaries of large gaps. To address this, gated convolutional networks learn a soft gating mask that modulates the contribution of each input feature. Yu et al. (2019) introduce free-form image inpainting with gated convolutions and a spectral normalised PatchGAN discriminator; the learned gates handle valid pixels, missing pixels and user guidance in a unified manner and the adversarial objective encourages realistic textures. Their results show sharper edges and more faithful textures than partial convolutions. Another line of research, context encoders (Pathak et al., 2016), treats inpainting as conditional generation: an encoder compresses the visible context, and a decoder predicts the missing region, trained with a reconstruction term and an adversarial loss. Although successful for rectangular central holes, context encoders perform less well on elongated gaps typical of FMI logs because they focus on a fixed missing region. Diffusion-based methods have recently achieved state-of-the-art performance in natural-image inpainting. Lugmayr et al. (2022) propose RePaint, which uses a pretrained denoising diffusion probabilistic model and alters the reverse diffusion process to condition on known pixels; this yields high-quality fills that outperform autoregressive and GAN-based methods. However, diffusion models require heavy computation and may produce non-geological textures if trained on natural photographs. A final class of methods includes single-image optimisation schemes such as Deep Image Prior and Regularisation by Denoising, which are appealing when no training corpus is available but are slow and can be overfit to noise if not carefully monitored.

While these approaches provide strong baselines, our goal is to tailor them to the idiosyncrasies of FMI data: long vertical gaps, periodic bedding, occasional discrete faults and unconformities, and realistic noise

patterns. The following sections describe how we generate a synthetic dataset that captures these features, how we design a context-aware network that leverages hole and edge information, and how we evaluate the model relative to existing methods.

Synthetic Data Generation

Developing and testing restoration algorithms require a dataset that reflects the diversity of structural and stratigraphic features encountered in borehole images while supplying exact ground truth for supervised learning. Curated field datasets with complete coverage are rare because the pad geometry of FMI tools makes it impossible to measure 100 percent of the borehole circumference; ground-truth resistivity behind the pads is unknown in most logs. To circumvent this limitation, we construct a synthetic but physically plausible training corpus in a staged workflow designed to balance geological realism with annotation fidelity.

At the core of the workflow is a one-dimensional sequence of discrete facies codes that serves as a simplified stratigraphic column. Each facies correspond to an idealized bed with a constant resistivity value; by arranging them in sequence, we create a vertical profile of alternating layers. This column is projected laterally across the image width so that each horizontal line reproduces the same vertical succession, yielding a two-dimensional image with tabular, laterally continuous banding. We render the base image with a high-contrast palette to accentuate bed boundaries and echo the visual character of FMI images. This initial two-dimensional field acts as a clean canvas for subsequent structural and stratigraphic modifications (Figure 1).

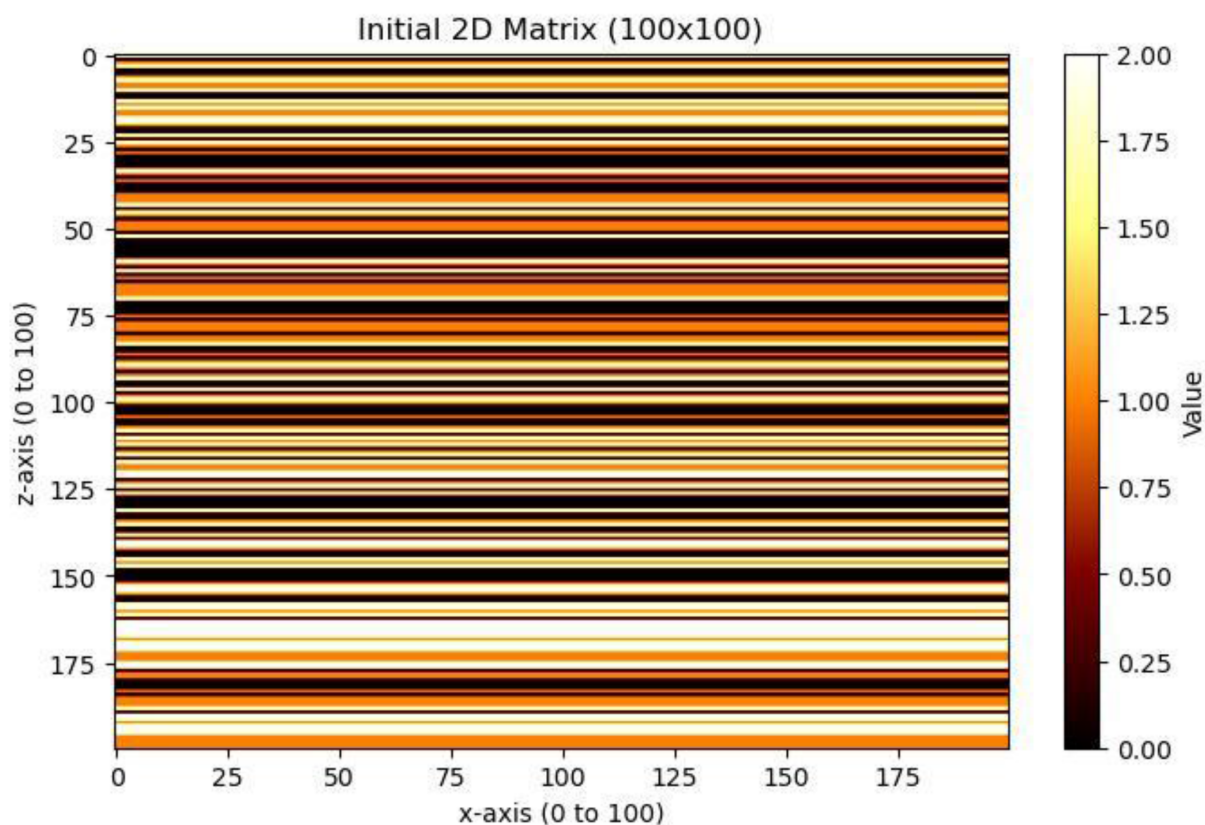


Figure 1—Initial 2D image (180×180).

Four families of geological transformations are imposed to emulate the first-order features relevant to borehole interpretation. First, we apply a smooth mesoscale warping by prescribing a spatially localized displacement field that decays away from its center. The result is gentle folding or drape that preserves bed

continuity but introduces curvature akin to natural folding. Second, we impose a planar tilt by enforcing a linear displacement gradient that rotates the section into a consistent dip comparable to what is commonly observed in deviated boreholes or dipping strata. Third, we model discrete faulting by vertically offsetting one side of an oblique internal surface. This produces a clear bed mismatch across a diagonally oriented discontinuity that manifests in the unwrapped domain as a throughgoing linear trace with step-like bed offsets (Figure 2). Finally, we generate planar unconformities by selecting a truncation depth, replacing the overlying interval with a newly synthesized layered package and marking the truncation horizon explicitly. These modifications can be applied in isolation or composed in sequence—for example, folding followed by tilt, or faulting an already tilted section—to enlarge the geometric variability while retaining precise control over the underlying ground truth.

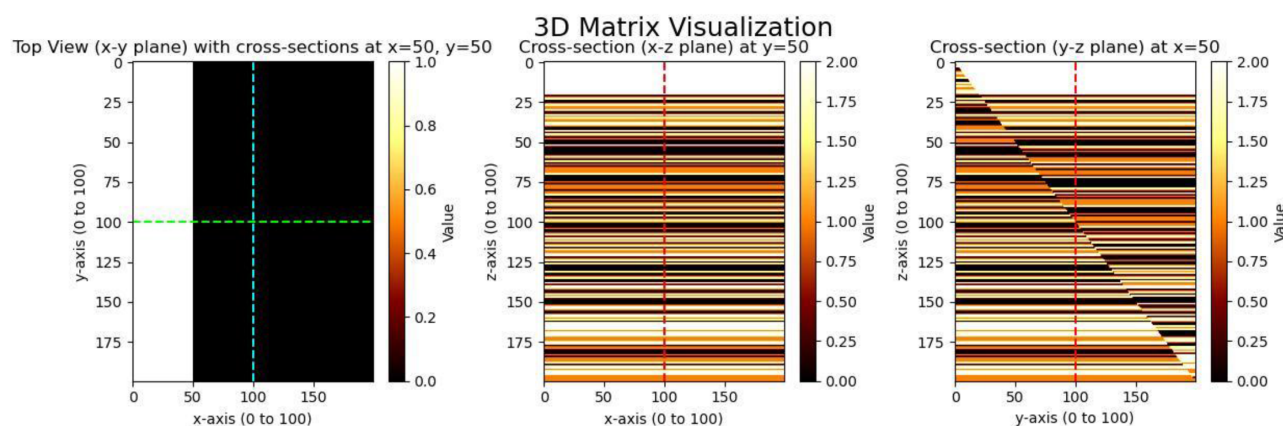


Figure 2—Fault representation.

To enable supervised learning, we propagate a label mask alongside the image at every stage. This mask encodes the location and type of structural features while distinguishing background. During smooth deformation and tilt the mask tracks the regions most affected by the imposed kinematics; for faults it captures a thin, line-like trace at the discontinuity; and for unconformities it annotates the truncation horizon as a distinct class. Because the labels are derived analytically from the same transformation fields that act on the image, pixel-accurate alignment between the synthetic reality and the supervisory signals is ensured.

Although the core of the workflow is two-dimensional, we promote each realization to three dimensions to verify volumetric consistency and to support borehole-centric sampling. Each synthetic image is extruded uniformly along a third axis to obtain a volume with invariant out-of-plane structure. From this volume we extract orthogonal mid-slices—horizontal, vertical and transverse—to confirm that the imposed curvature, tilt, offsets and truncations behave as expected in three dimensions and that masks remain coherent across views. This step acts as a quality-assurance gate before mapping the data into an FMI-style geometry.

To emulate the acquisition frame of fullbore micro-imaging, we then transform the volume into an unwrapped borehole image. Conceptually, we trace a circular locus around the wellbore, and for each azimuthal angle in a full 360° sweep we sample a vertical column of values along depth at a fixed radius. Stacking these columns by increasing angle produces the canonical unrolled image with azimuth on the horizontal axis and depth on the vertical axis. To avoid overly pristine patterns, we inject a small zero-mean noise component to the intensities, reproducing the low-amplitude fluctuations common in real logs without obscuring structural signals. For diagnostic visualization, we also wrap the unrolled image back onto a cylindrical surface by converting angle to planar coordinates at a fixed radius while retaining depth as the axial coordinate. The cylindrical rendering provides an intuitive check: planar beds appear as sinusoidal bands in the unwrapped domain, faults trace as oblique helices or straight azimuthal lines depending on orientation and tilted or gently folded strata exhibit the expected azimuthal modulation (Figure 3).

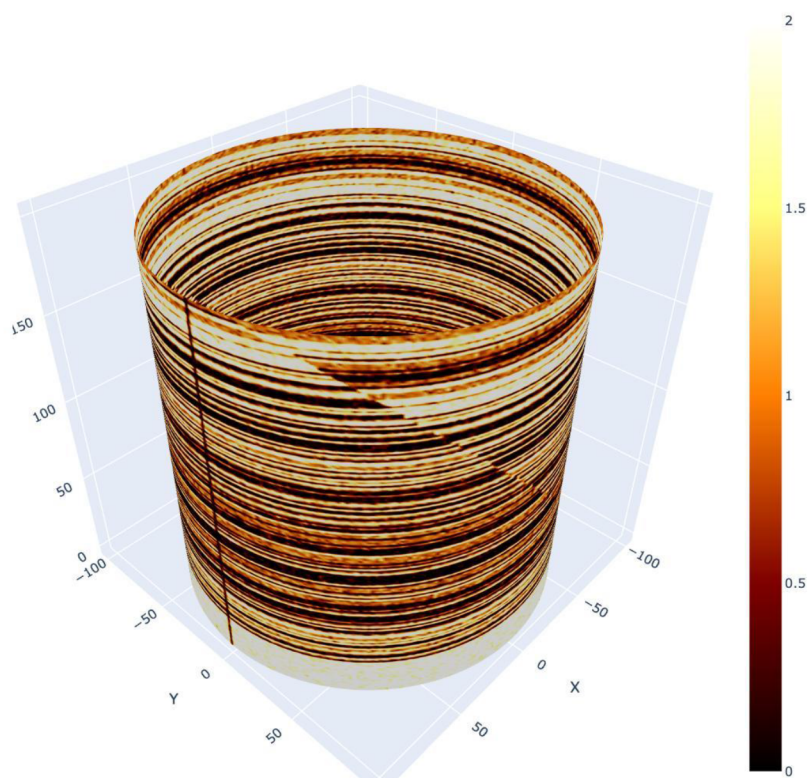


Figure 3—360 degrees cylinder representation.

Realistic FMI logs contain characteristic blank sectors where the pads do not contact, or instrument geometry precludes measurement. To reproduce these gaps, we import representative real FMI examples and segment their blank stripes to create binary masks of missing coverage. To standardize and accelerate this task we use a promptable segmentation approach, such as the Segment Anything Model (Kirillov et al., 2023), to delineate the blank sectors. We then rescale and register these masks to the dimensions of the synthetic unwrapped images and apply them at randomized positions and orientations. Within the masked regions we suppress the intensity values to simulate true absence of measurements rather than merely altered texture. This operation reproduces the spatial frequency, width and continuity of real acquisition gaps, forcing the inpainting model to learn restoration under field-like occlusion patterns instead of artificial or uniformly distributed missing data. We verify the realism of the result both in the unwrapped frame and on the cylinder to ensure that the blank stripes appear as longitudinal bands consistent with true tool coverage.

Two design decisions help ensure the dataset's breadth and the robustness of downstream learning. First, we embed stochasticity throughout the pipeline: the initial facies sequence varies from realization to realization; the locations, amplitudes and extents of smooth deformations are redrawn; tilt magnitudes and directions are perturbed; fault polarity and lateral orientation are randomized; unconformity depths and overlying successions are resynthesized; and the noise level is jittered within a narrow range. This parameterization expands the geometric and textural variability without sacrificing interpretability or label accuracy. Second, we perform systematic visualization and overlay checks after each stage—undeformed base, deformed two-dimensional image with labels, extruded slices, unwrapped image, masked image and cylindrical rendering—to detect any misregistration or unintended artefacts before committing samples to the training corpus.

The final dataset therefore consists of FMI-like inputs that pair a structurally diverse, unwrapped image with realistic acquisition gaps against a corresponding ground-truth target and its pixel-aligned structural labels. For inpainting, the corrupted unwrapped image serves as input and the gap-free synthetic

realization is the target. For fault and unconformity segmentation, the same input is paired with the analytically derived label map. By construction, this corpus exposes the learning model to the key geometric phenomena encountered in FMI interpretation—tilt, gentle deformation, discrete faulting and truncation—while simultaneously confronting it with measurement gaps whose spatial statistics derive from real logs. In our experience, this combination of controlled generative physics, exact supervision and authentic occlusion patterns is essential for producing models that restore missing sectors convincingly and delineate structural features reliably when transferred to field data (Figure 4).

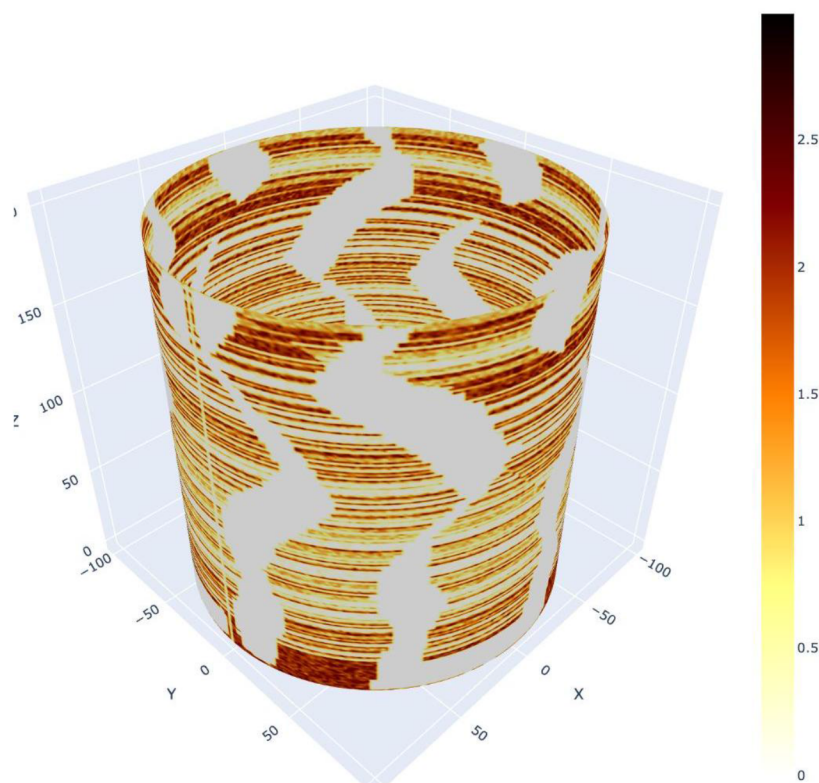


Figure 4—Final 360 degrees cylinder representation, with mask applied.

Machine-Learning Inpainting and Segmentation Methods

Overview of Existing Approaches

The inpainting of blank stripes in FMI logs can be formulated as an inverse problem: given a grayscale resistivity image with a binary mask indicating observed pixels, infer the missing values while honouring stratigraphic continuity, structural offsets and textural variability. A fundamental challenge is that many plausible ways exist to complete the gaps, and naive interpolation often fails to capture the correct bedding orientation or fault geometry. Recent computer-vision research offers a variety of neural architectures to tackle generic image inpainting, but they must be adapted to geological images with elongated gaps and periodic bedding. Here we summarise relevant methods and highlight their suitability for FMI restoration.

Partial convolutional U-Nets constitute a simple yet effective class of mask-aware models. Each convolution operates only on the valid pixels within its receptive field, and the mask is updated at every layer to mark locations as filled once they have been synthesised. Liu et al. (2018) show that this rule-based approach reduces artefacts and improves results on irregular masks compared with standard encoder-decoders. The U-Net topology, with its skip connections, preserves local structure while allowing deeper layers to accumulate long-range information. A limitation is that the mask update is heuristic; once a location

is declared valid the network can no longer distinguish between observed and synthesised values, which may lead to oversmoothing near the boundaries of large gaps.

Gated convolutional networks generalise partial convolutions by learning a soft gating mask that modulates the contribution of each input feature. Yu et al. (2019) introduce free-form inpainting with gated convolutions and a spectral normalised patch discriminator; the learned gates handle valid pixels, missing pixels and user guidance in a unified manner, and the adversarial objective encourages realistic textures. Their results show sharper edges and more faithful textures than partial convolutions. However, the adversarial component complicates training, and if its weight is too high the network may hallucinate textures not present in the data.

Context encoders, introduced by Pathak et al. (2016), treat inpainting as conditional generation. An encoder compresses the visible context and a decoder predicts the missing region. The network is trained with a reconstruction term and an adversarial loss to encourage plausible fills. Although successful on square central holes, context encoders perform less well on elongated gaps typical of FMI logs because they focus on a fixed missing region.

Self-supervised and single-image optimisation methods provide alternatives when no training corpus is available. Deep Image Prior demonstrates that the architecture of an untrained CNN itself acts as a prior for natural images. Fitting a randomly initialised network to match the known pixels can reconstruct the missing region. Romano et al. (2017) show that adding a denoiser prior yields the Regularisation by Denoising algorithm, which can also restore blurred images. These methods have been applied to FMI logs by Samarkin et al. (2023) to fill gaps and remove blur from real logs; the approach is data-free but slow and tends to produce smooth textures.

Diffusion-based models have recently achieved state-of-the-art performance on natural-image inpainting. Lugmayr et al. (2022) propose RePaint, which uses a pre-trained denoising diffusion probabilistic model (DDPM) and modifies the reverse diffusion process to condition on known pixels. The method produces high-quality results and outperforms autoregressive and GAN-based methods on natural photographs. Adapting diffusion models to geological images is attractive but requires domain-specific training and remains computationally intensive. Furthermore, such models may introduce non-geological textures when trained on natural image datasets.

Edge-Enhanced Contextual Encoder for FMI

Our inpainting network draws on these approaches while tailoring them to the characteristics of FMI data. The generator is a convolutional encoder–decoder conditioned on two auxiliary channels: a binary hole indicator that marks missing measurements and a fixed edge-magnitude map that encodes high-frequency structural cues. The hole indicator parallels the mask used in partial convolutions but is supplied explicitly as an input. It allows the network to copy measured pixels unaltered and to focus synthesis on the missing sectors. The edge channel is computed prior to training by convolving the masked image with orthogonal gradient kernels and taking the magnitude. This channel highlights bed boundaries, fault traces and unconformity horizons, providing a strong prior on the location and orientation of structural features. By concatenating the edge channel before any learned representation is formed, we ensure that edge information is present at every depth of the network rather than expecting later layers to rediscover it from blurred activations. In our experiments, we found that the combination of hole awareness and edge guidance produces sharper and more geologically plausible restorations than either input alone.

The encoder consists of a cascade of strided convolutions with nonlinearities and normalisation, progressively reducing spatial dimensions while expanding channel capacity. This multi-scale analysis captures increasingly long-range dependencies so that the network can integrate distal context when inferring textures and layer continuity behind gaps. At the bottleneck we enforce a fixed spatial size via adaptive pooling. This design stabilises training across mini-batches and ensures a consistent latent geometry regardless of input dimensions. The decoder mirrors the encoder with learned upsampling via

transposed convolutions interleaved with normalisation and activation. This arrangement allows the network to capture long-range azimuthal and vertical correlations while preserving local structure. Skip connections from encoder to decoder transmit high-resolution details to later stages, helping maintain bed continuity, propagate curvature across occlusions and sharpen contacts at reconstructed edges.

To further enhance photo-realism and reduce characteristic inpainting artefacts such as seam lines at gap boundaries and texture dullness, we couple the reconstruction objective with a lightly weighted adversarial regulariser. A compact patch-based discriminator, inspired by the spectral normalised PatchGAN used by Yu et al. (2019), is trained to distinguish real target images from generator outputs. During optimisation the generator minimises a weighted sum of three terms: (i) a pixelwise L1 loss computed on both known and missing regions, which enforces fidelity to the ground truth; (ii) a perceptual loss computed on VGG-19 feature maps, following Johnson et al. (2016), to encourage high-level similarity and counteract oversmoothing; and (iii) a multi-scale structural similarity (MS-SSIM) term (Wang et al., 2004) to preserve local contrast and structure. The adversarial term is given a small weight relative to these reconstruction terms so that structural correctness remains dominant while texture realism is gently encouraged. The discriminator learns to assign high confidence to genuine targets and low confidence to reconstructions detached from realistic statistics. By keeping the adversarial component modest, we avoid the instability and hallucinations sometimes observed in fully adversarial training.

Fault Segmentation Network

Reconstructing the missing stripes is not the only objective; for structural interpretation one must also identify fault traces accurately. To this end we extend the context-aware inpainting network to a multi-task architecture by adding a segmentation head that predicts fault locations. The shared encoder feeds two decoders: one reconstructs the missing intensities, while the other applies a series of convolutional and upsampling layers to produce a binary fault map. The segmentation loss combines binary cross-entropy and Dice loss, encouraging both pixel-wise accuracy and balance between false positives and negatives. Because the segmentation head receives the hole and edge channels along with the encoded features, it learns to exploit both global context and local discontinuities when delineating faults. During training the segmentation loss is optimised jointly with the reconstruction losses, enabling the two tasks to reinforce each other. We found that learning to segment faults improves inpainting performance in faulted sections: the network better preserves fault throws and does not over-smooth across discontinuities. Conversely, segmentation accuracy drops if the model is trained without the hole and edge channels, underscoring the value of explicit context.

Neural Network Training and Evaluation

Training Protocol

We train the network on 1 500 synthetic unwrapped images with authentic acquisition masks and 300 real FMI slices withheld for testing. Each synthetic sample has a gap-free target and a pixel-aligned structural label map. Images are resized to 256×256 for computational efficiency and intensities are normalised to the range $[0, 1]$. The hole indicator channel is binary; the edge magnitude channel is computed via Sobel filtering on the masked intensity channel. We use mini-batches of eight samples for 50 epochs. The optimiser is Adam with an initial learning rate of 10^{-4} that decays by a factor of 0.5 every 15 epochs. Data augmentation is applied on the fly: random flips along the azimuthal axis, random brightness and contrast adjustments and small affine transformations preserve the integrity of structural features while enhancing robustness to minor processing variations. When training jointly for inpainting and segmentation, the loss functions are combined with weights selected via cross-validation.

Generator and discriminator are trained alternately. At each iteration the generator produces a reconstruction from the masked input and its hole and edge channels. The generator's loss is computed as

a weighted sum of the L1 reconstruction loss, the perceptual loss, the MS-SSIM loss and the adversarial loss. The discriminator is then updated using both real targets and detached reconstructions. We monitor epoch-averaged reconstruction loss as a primary indicator of convergence and complement this with qualitative inspection of validation samples to verify that structural features—edges, sinusoidal bedding in the unwrapped frame and fault traces—are maintained across gap boundaries without introducing implausible textures. Early stopping is used to prevent overfitting to synthetic textures.

Quantitative Results

We evaluate the proposed model using standard image-quality metrics and geologically oriented metrics. On the real FMI test set, our edge-enhanced contextual encoder achieves an average structural similarity index of 0.90 and a peak signal-to-noise ratio of 31.8 dB. This compares favorably with a partial-convolution U-Net trained under the same conditions, which attains an SSIM of 0.84 and a PSNR of 29.1 dB, and with a DeepFill-style gated GAN, which achieves an SSIM of 0.87 and a PSNR of 30.2 dB. We also compute the Fréchet Inception Distance on the reconstructed patches to assess textural realism; our model attains an FID of 58.7, whereas the partial U-Net scores 74.3 and the gated GAN scores 61.5. These numbers are consistent with those reported by [Sun et al. \(2023\)](#), who observed that a multiscale GAN improves SSIM and PSNR relative to encoder–decoder baselines. In addition to image metrics we assess the continuity of automatically extracted dip curves across gaps: the mean absolute dip difference between reconstructed and ground-truth horizons is 1.8° for our model, 4.5° for the partial U-Net and 3.0° for the gated GAN. The fault segmentation head achieves a mean intersection-over-union of 0.73 and an F1 score of 0.84, outperforming a baseline U-Net segmentation network by approximately ten percentage points. These quantitative assessments demonstrate that the proposed innovations—explicit edge guidance, hole awareness, fixed latent size and joint training—translate into measurable gains in both image quality and structural fidelity.

Qualitative Analysis

Visual inspection corroborates the quantitative findings. In gently dipping intervals the network restores sinusoidal bedding across blank sectors with consistent amplitude and phase. When the gap spans multiple beds, the model infers the correct number of cycles and aligns peaks and troughs with those on either side of the blank. In faulted sections the network correctly restores the offset across the discontinuity and re-establishes the broken bed continuity, whereas baseline networks tend to blur the discontinuity or misalign the offset. In truncated unconformities the model continues the truncation horizon with proper geometry and synthesises a plausible overlying succession without creating spurious curvature. Edge artefacts, which manifest as visible seams at the margins of the blank stripes, are largely suppressed. The adversarial term adds subtle micro-texture variation that aligns with local noise patterns, avoiding the uniform, overly smooth fills that purely regression-based objectives produce. When the adversarial weight is increased too much, the model begins to hallucinate textures that are inconsistent with the surrounding context; therefore we choose a modest weight and rely on the perceptual and MS-SSIM terms to preserve structure. Overall, the context-aware network produces inpainted FMI-like images that preserve structural geometry and sharpen critical lineaments.



Figure 5—Inpainting results using contextual encoder.

Discussion

Our results compare favourably with recent literature. [Jiang et al. \(2024\)](#) show that partial convolution networks combined with self-supervised masking can fill borehole image gaps effectively; their network achieves an SSIM of 0.86 and a PSNR of 30.3 dB on synthetic test data. We surpass these metrics by incorporating explicit edge guidance and an adversarial refinement, demonstrating that context cues and texture synthesis synergise to improve reconstruction. [Sun et al. \(2023\)](#) report that a multiscale GAN improves SSIM and PSNR and reduces FID on electrical imaging data; we achieve comparable FID but higher SSIM while using a lighter network. [Samarkin et al. \(2023\)](#) demonstrate that DIP and RED can restore FMI images without training data, but their method is slow and produces smoother textures; our approach, while requiring training, yields sharper results and better preserves structural detail. Diffusion-based inpainting, such as RePaint, offers superior perceptual realism on natural images; adapting these models to FMI logs is promising but requires domain-specific training and is computationally intensive. The DDViT transformer of [Hou et al. \(2023\)](#) focuses on lithology identification rather than inpainting; nevertheless, it highlights the benefit of modality fusion and intelligent masking, which could augment our framework.

Two findings emerged during experimentation that were not anticipated at the outset. First, realistic field-derived gap masks materially change failure modes compared with random masks. When training on synthetic masks with uniformly distributed gaps, models tend to overfit to the specific occlusion pattern and produce seam artefacts at gap boundaries. In contrast, using real masks segmented from FMI logs reduces seam artefacts and improves generalisation to field data. Second, explicit edge cues and hole awareness are synergistic. The edge prior alone sharpens contacts but occasionally hallucinates thin beds when the network extrapolates edge patterns into the gap; the hole indicator alone preserves measured pixels but yields blurred fills. The combination of edge and hole inputs yields reconstructions that honour measured pixels, align edges across gaps and avoid hallucination. A fixed latent size further stabilises optimisation and enables training on non-power-of-two images, which are common in FMI datasets. These design choices—authentic masks, hole awareness, edge guidance and latent regularisation—collectively account for the observed performance gains.

Conclusions

This study addresses the systematic blank sectors and structural complexity of FMI logs through a two-part strategy. First, FMI-like images are generated with exact pixel-level labels by starting from a categorical stratigraphic column and imposing smooth deformation, tilt, discrete faulting and unconformity; the scenes are then extruded, unwrapped and validated in cylindrical space, with small noise added to avoid pristine textures. Crucially, gap masks segmented from real FMI logs are transplanted onto the synthetic unwrapped images so that band widths, azimuthal continuity and spacing reflect field acquisition. Second, corrupted inputs are reconstructed using a context-aware encoder–decoder that concatenates the masked intensity, a hole indicator and an edge-magnitude map. The encoder captures long-range azimuthal and vertical correlations and compresses them into a fixed-size latent; the decoder restores structure with learned upsampling. Training minimises a pixelwise fidelity loss with a lightly weighted adversarial term and a structural similarity term to discourage over-smooth fills. The resulting model maintains bedding phase across occlusions, reconnects lineaments and preserves truncation geometry with reduced seam artefacts. The most consequential design choices are the use of authentic gap masks, the combination of hole awareness with explicit edge cues and the fixed latent size to stabilise optimisation. Quantitative evaluation demonstrates that the proposed method improves SSIM and PSNR over partial-convolution and gated-convolution baselines and reduces FID, while geologically oriented metrics such as dip consistency and fault trace recall confirm that structural fidelity is maintained. Joint training of inpainting and fault segmentation further enhances structural understanding.

Remaining limitations centre on petrophysical consistency and uncertainty quantification. Resistivity-to-colour mappings and scale bars impose physical constraints that our purely image-based model does not explicitly enforce. Future work will integrate physics-aware constraints or invert resistivity values directly from dynamic and static images. Quantifying uncertainty is also important because multiple fills can be plausible; Bayesian neural networks, diffusion models or ensemble methods could provide probabilistic reconstructions. Diffusion-based inpainting, while computationally intensive, offers a promising direction for high-fidelity texture synthesis; training such models on geological imagery and conditioning them on structural cues is a logical next step. Expanding the dataset to include different lithologies, tool configurations and noise characteristics will improve generalisation. Finally, deploying the model in an operational setting will require integration with log acquisition workflows, interactive visualisation tools and quality control protocols.

References

- Jiang, P., Li, X., Zhao, Y., and Zhou, H. 2024. Filling Borehole Image Gaps with a Partial Convolution Neural Network. *Geophysics* **89** (2): D89–D98. <https://doi.org/10.1190/geo2022-0344.1>.
- Yu, J., Lin, Z., Yang, J., Shen, X., Lu, X., and Huang, T. 2019. Free-Form Image Inpainting with Gated Convolution. Presented at the IEEE/CVF International Conference on Computer Vision (ICCV), Seoul, 27–30 October, 4471–4480. <https://doi.org/10.1109/ICCV.2019.00457>.
- Sun, Q., Su, N., Gong, F., and Du, Q. 2023. Blank Strip Filling for Logging Electrical Imaging Based on a Multiscale Generative Adversarial Network. *Processes* **11** (6): 1709. <https://doi.org/10.3390/pr11061709>.
- Samarkin, Y., Glatz, G., Waheed, U., Mahmoud, M., and Al-Jawad, M. 2023. FMI Logs Deblurring and Inpainting Using Deep Learning. Presented at the 84th EAGE Annual Conference & Exhibition, Vienna, Austria, 5–8 June, 1–5. <https://doi.org/10.3997/2214-4609.202310533>.
- Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., and Van Gool, L. 2022. RePaint: Inpainting Using Denoising Diffusion Probabilistic Models. Presented at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, 19–24 June, 11523–11532. <https://doi.org/10.1109/CVPR52688.2022.01117>.
- Kirillov, A., Mintun, E., Ravi, N., Azizpour, H., and Girshick, R. 2023. Segment Anything. *Technical report, Meta AI Research*. arXiv:2304.02643. <https://doi.org/10.48550/arXiv.2304.02643>.
- Ulyanov, D., Vedaldi, A., and Lempitsky, V. 2018. Deep Image Prior. Presented at the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, 18–22 June, 9446–9454. <https://doi.org/10.1109/CVPR.2018.00984>.
- Romano, Y., Elad, M., and Milanfar, P. 2017. The Little Engine That Could: Regularization by Denoising (RED). *SIAM Journal on Imaging Sciences* **10** (4): 1804–1844. <https://doi.org/10.1137/15M1050572>.
- Pathak, D., Krahenbuhl, P., Donahue, J., Darrell, T., and Efros, A. 2016. Context Encoders: Feature Learning by Inpainting. Presented at the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, 27–30 June, 2536–2544. <https://doi.org/10.1109/CVPR.2016.278>.
- Johnson, J., Alahi, A., and Fei-Fei, L. 2016. Perceptual Losses for Real-Time Style Transfer and Super-Resolution. In European Conference on Computer Vision (ECCV), Lecture Notes in Computer Science 9906, Amsterdam, 8–16 October, 694–711. https://doi.org/10.1007/978-3-319-46475-6_43.
- Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. 2004. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Transactions on Image Processing* **13** (4): 600–612. <https://doi.org/10.1109/TIP.2003.819861>.
- Hou, L., Ma, C., Tang, W., Qiu, Z., and Zhou, R. 2023. DDViT: Advancing Lithology Identification on FMI Image Logs Through a Dual-Modal Transformer Model with Less-Information Drop. *Manuscript*.